

Theoretical Constraints on Neutron Star Superfluidity from Her X-1 Precession

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ABSTRACT

Recent IXPE observations of Her X-1 reveal correlations between flux, polarization degree, and polarization angle with its 35-day superorbital cycle. These measurements have been interpreted as strong evidence that the 35-day period is driven by nearly free precession of the neutron star. We show that this interpretation carries far-reaching implications for the dynamics of the crustal superfluid. In particular, maintaining precession over the ~ 50 -year observational baseline of Her X-1 would require that superfluid vortices remain unpinned for centuries and experience extremely weak mutual friction while traversing the heavy-ion lattice of the inner crust – conditions that challenge conventional wisdom and standard models of glitch dynamics.

1. INTRODUCTION

Neutron stars are born hot but cool rapidly, and a substantial fraction of their interior is expected to transition into a superfluid state within years to decades after formation. In particular, neutrons in the inner crust and outer core are widely believed to form a superfluid capable of sustaining long-lived, weakly dissipative flows on macroscopic scales. In a rotating neutron star, the angular momentum of this superfluid is carried by an array of quantized vortex lines (G. Baym et al. 1969), whose areal density is directly proportional to the coarse-grained angular velocity of the superfluid (I. M. Khalatnikov 1965; G. Baym & E. Chandler 1983).

It has long been thought that pulsar glitches – sudden discrete increases in the observed rotation frequency – provide some of the most direct evidence for the presence of a superfluid component in neutron stars (for reviews, see B. Haskell & A. Melatos 2015; M. Antonelli et al. 2022, and references therein). In the standard picture, vortices in the crustal superfluid are partially pinned to the nuclear lattice, allowing the superfluid to rotate slightly faster than the solid crust as the star spins down under electromagnetic torques. Differential rotation then builds up until a critical threshold is reached, triggering rapid vortex unpinning and outward motion (P. W. Anderson & N. Itoh 1975; M. A. Alpar et al. 1984b,a; B. Link 2014). The resulting transfer of angular momentum from the superfluid to the crust produces the observed glitch.

Following a glitch, the star typically exhibits a relaxation phase, during which the rotation rate evolves toward a new equilibrium on timescales ranging from days to months. This post-glitch recovery is commonly interpreted as the gradual re-coupling of superfluid components that were only weakly coupled to the crust during the glitch itself. In particular, mutual interactions between vortices and the surrounding charged fluid mediate the exchange of angular momentum between the crust and the superfluid in the core, leading to partial or complete recovery of the pre-glitch spin-down rate. The detailed form of the relaxation thus encodes information about the strength of superfluid coupling and the distribution of superfluid reservoirs within the star.

Vortex pinning can have a profound effect on the free precession of a neutron star by dynamically coupling the superfluid component to the solid crust. In the inner crust, quantized vortices of the neutron superfluid are expected to pin to the nuclear lattice, preventing the superfluid from freely adjusting its angular momentum in response to changes in the crust’s rotation. If vortices remain strongly pinned during precession, the pinned superfluid exerts a dominant gyroscopic restoring force on the crust that controls the crust’s precessional dynamics. As originally shown by J. Shaham (1977) (see also D. I. Jones & N. Andersson 2001; B. Link & C. Cutler 2002), in case of absolute pinning the precessional angular frequency must be of the order $\omega_p \sim (I_s/I)\Omega_s \gtrsim 10^{-2}\Omega$, where I_s/I is the fraction of the stellar moment of inertia in the pinned superfluid and Ω is the angular frequency of the star.

The fast precessional periods (a few minutes typically) imposed by absolute pinning are inconsistent with the

much longer precessional periods inferred for some neutron stars, most notably Hercules X-1. It has long been suspected that the observed 35-day superorbital period in Hercules X-1 may be due to (nearly free) precession of the neutron star. Recent X-ray polarization measurements with the Imaging X-ray Polarimetry Explorer (IXPE), which can directly probe the spin geometry of the neutron star, have recently been argued to support this interpretation (J. Heyl et al. 2024; Q. C. Zhao et al. 2024).

A. Alpar & H. Oegelman (1987) proposed that a steady state may exist in which a pinned superfluid follows the precessional motion of the crust through thermal vortex creep. In this picture, the component of the superfluid–crust lag perpendicular to the vortex lines is of order the precession frequency ω_p and therefore much smaller than the critical lag required for unpinning. The relaxation time to this equilibrium depends on the ratio of the pinning energy to the thermal energy and was estimated to be of the order of a few hundred to a few thousand years for Her X-1. However, A. Alpar & H. Oegelman (1987) neglected the non-dissipative torque associated with the component of the mutual-friction force acting in the direction $\tilde{\omega} \times (\mathbf{v}_n - \mathbf{v}_s)$, where $\mathbf{v}_n$ and $\mathbf{v}_s$ are the normal-fluid and superfluid velocities, respectively, and $\tilde{\omega}$ is the superfluid vorticity. As we shall show, this omission is not justified.

The torque associated with the mutual-friction force was derived rigorously by A. Sedrakian et al. (1999) and applied to the calculation of precessional modes in the case of imperfect pinning. In the strong-drag regime, they found low-frequency, strongly damped free-precession modes and concluded that their results are qualitatively consistent with J. Shaham (1977), in that no long-lived free precession survives in the strong-drag regime. However, their analysis is perturbative and restricted to oblate deformations, and considers only small (linear) lags about a fixed point in which all angular velocities are equal and aligned with a stable principal axis. It therefore remains unclear how the rotational dynamics behaves in the general case.

In this paper, we present a comprehensive analysis of the rotational evolution, taking into account the internal torque exerted on the normal (rigid) crust by the crustal superfluid and assuming tight coupling between the core and the normal crust. We apply these results to Her X-1 and show that if the 35-d cycle is produced by nearly free precession of the neutron star, then superfluid vortices must remain unpinned for centuries or longer and experience extremely weak mutual friction, with a dimensionless drag coefficient $\lesssim 0.01$, while traversing the heavy-ion lattice of the inner crust. This implies that the

existence of long-lived precession is incompatible with the buildup of the large differential rotation required to trigger glitches.

2. TWO-COMPONENT MODEL

Three main components govern a neutron star’s rotational dynamics: the outer core, the elastic charged lattice in the crust, and the crustal neutron superfluid. In isolated neutron stars, the core temperature is expected to fall below the neutron superfluid transition temperature, $T_n \sim 5 \times 10^8$ K, so that neutrons in the core become superfluid. In accreting neutron stars, however, the core temperature may approach T_n (E. F. Brown 2000; but see M. Nava-Callejas et al. 2025), allowing neutrons in much of the outer core to remain in a normal state (Wasserman, private communication). In both cases, the temperature is well below the critical temperature for proton pairing, so that protons form either a Type I or Type II superconductor. B. Link (2003) argued that if the core contains coexisting superfluid vortices and superconducting proton flux tubes, pinning between vortices and flux tubes would suppress long-period precession, unless vortices can somehow cut through the flux tubes. Here, we ignore pinning of vortices to flux tubes (as expected, e.g., when the neutrons in the core are normal or the protons are in an intermediate state). Under these conditions, the core–crust coupling is mediated by electromagnetic forces, with a strength that depends on the detailed microphysics (see, e.g., Y. Levin & C. D’Angelo 2004 for details). Here, we adopt, for illustration, the simple phenomenological model of A. Alpar & H. Oegelman (1987) in which the torque is given by:

$$\mathbf{N}_{co} = I_{co} \tau_{co}^{-1} (\mathbf{\Omega}_{co} - \mathbf{\Omega}_n), \quad (1)$$

where the relaxation time τ_{co} depends on the specific coupling mechanism. Here I_{co} and $\mathbf{\Omega}_{co}$ are, respectively, the moment of inertia and angular velocity vector of the core, and $\mathbf{\Omega}_n$ is the angular velocity vector of the elastic (normal) crust. In the case of electron scattering off magnetized superfluid vortices M. A. Alpar & J. A. Sauls (1988) obtained

$$\tau_{co} \sim 400 - 10^4 P_n \quad (2)$$

where $P_n = 2\pi/\Omega_n$ is the rotation period, about 1.2 sec in Her X-1. In case where the neutrons in the core are normal, the coupling time depends on the strength and geometry of the magnetic field in the core. In Appendix A, we show that, for a deformed crust with ellipticity ϵ_{cr} and an isotropic core, free precession is damped on a timescale

$$\tau_p \approx \frac{I_n^2 (\Omega_n \tau_{co})^2 + I_{co}^2}{\epsilon_{cr} I_{co} I_n \Omega_n^2 \tau_{co}}, \quad (3)$$

where I_{co} and I_{n} are the moments of inertia of the core and crust, respectively. As expected, $\tau_{\text{p}} \rightarrow \infty$ both in the limit $\tau_{\text{co}} \rightarrow \infty$ (no coupling) and $\tau_{\text{co}} \rightarrow 0$ (infinitely tight coupling). Using Eq. (2), we find that $\Omega_{\text{n}}\tau_{\text{d}} \gg I_{\text{co}}/I_{\text{n}}$ for a fiducial ratio $I_{\text{co}}/I_{\text{n}} \approx 10^2$. In this regime

$$\tau_{\text{p}} \approx \epsilon_{\text{cr}}^{-1} \left(\frac{I_{\text{n}}}{I_{\text{co}}} \right) \tau_{\text{co}} \approx 3 \times 10^2 \tau_{\text{co}}, \quad (4)$$

adopting $\epsilon_{\text{cr}} = (\Omega_{\text{p}}/\Omega_{\text{n}})(I_{\text{co}}/I_{\text{n}}) = 3 \times 10^{-5}$ for a precession frequency of $\Omega_{\text{p}} = 2\pi/35$ days, as inferred from observations of Her X-1. For $\tau_{\text{co}} \sim 10^4 P_{\text{n}}$, this gives $\tau_{\text{p}} \sim 0.1$ yr, shorter than the observational span (~ 50 yr). On the other hand, a deformed core will naturally induce deformations in the overlying crust. During precession, the crust continuously slips against the precessing core, a process expected to cause precession damping. A detailed analysis of the full core–crust interaction is beyond the scope of this work. Here, to obtain a rough estimate of the damping timescale, we assume an isotropic moment of inertia for the crust. Under this simplifying assumption one finds

$$\tau_{\text{p}} \approx \epsilon_{\text{co}}^{-1} \left(\frac{I_{\text{co}}}{I_{\text{n}}} \right) \tau_{\text{co}} \approx 3 \times 10^8 \tau_{\text{co}} \approx 10^5 \text{ yr}, \quad (5)$$

here $\epsilon_{\text{co}} = \Omega_{\text{p}}/\Omega_{\text{n}} \approx 3 \times 10^{-7}$ is the ellipticity of the core, indicating that the core and solid crust can be treated as a single rigid body on observational timescales (see also Y. Levin & C. D’Angelo 2004).

The deformation of the core may be caused by magnetic stresses. If the protons in the core form a type-II superconductor, the resulting ellipticity can be written as (C. Cutler 2002)

$$\epsilon_{\text{co}} = \frac{25r_{\text{NS}}^4}{24GM_{\text{NS}}^2} (2\langle B_{\text{p}}H_{\text{c1}} \rangle - \langle B_{\text{t}}H_{\text{c1}} \rangle), \quad (6)$$

for $B_{\text{p}}, B_{\text{t}} \ll H_{\text{c1}}$, where B_{p} and B_{t} are the poloidal and toroidal magnetic-field components in the core, angle brackets mean volume averaging over the interior, and $H_{\text{c1}} \approx 10^{15}$ G is the lower critical field for the superconducting transition. Thus, a toroidal field $B_{\text{t}} \sim 30B_{\text{p}} \approx 10^{14}$ G is sufficient to produce the required ellipticity, assuming $B_{\text{p}} \approx 4 \times 10^{12}$ G – roughly the measured poloidal field near the stellar surface in Her X-1. In this case pinning between vortices and flux tubes may be avoided if the neutrons are normal.

To simplify the analysis, we treat the core and the elastic crust as a single deformed rigid body. This is an optimal-case assumption, since any friction between the core and the crust would only increase the damping of precession. A fully self-consistent treatment incorporating all three components is deferred to future work.

2.1. Mutual-friction torques in the crust

The coupling between the crustal superfluid and the elastic crust is mediated by interactions between quantized vortices and crustal nuclei. This interaction is commonly modeled by a phenomenological drag force per unit length of a vortex: $\mathbf{f}_{\text{d}} = -\eta(\mathbf{v}_{\text{L}} - \mathbf{v}_{\text{n}})$, where η is a drag coefficient, $\mathbf{v}_{\text{L}}$ is the vortex velocity and $\mathbf{v}_{\text{n}} = \Omega_{\text{n}} \times \mathbf{r}$ is the velocity of the normal crustal component. The motion of a vortex is determined by balancing this drag force with the Magnus force arising from the relative motion between the bulk superfluid and the vortex line, $\mathbf{f}_{\text{M}} = \rho_{\text{s}}\kappa\hat{\omega} \times (\mathbf{v}_{\text{L}} - \mathbf{v}_{\text{s}})$, which leads to

$$\rho_{\text{s}}\kappa\hat{\omega} \times (\mathbf{v}_{\text{L}} - \mathbf{v}_{\text{s}}) = \eta(\mathbf{v}_{\text{L}} - \mathbf{v}_{\text{n}}). \quad (7)$$

Here $\kappa = h/2m_{\text{n}}$ is the quantum of circulation, $\hat{\omega}$ is a unit vector in the direction of local vorticity, ρ_{s} is the superfluid mass density, and $\mathbf{v}_{\text{s}}$ is its local velocity. The latter equation can be solved to yield

$$\begin{aligned} \mathbf{v}_{\text{L}} - \mathbf{v}_{\text{n}} &= \frac{\mathcal{R}}{1 + \mathcal{R}^2} \hat{\omega} \times (\mathbf{v}_{\text{n}} - \mathbf{v}_{\text{s}}) \\ &+ \frac{1}{1 + \mathcal{R}^2} \hat{\omega} \times (\hat{\omega} \times (\mathbf{v}_{\text{n}} - \mathbf{v}_{\text{s}})), \end{aligned} \quad (8)$$

where $\mathcal{R} = \eta/\rho_{\text{s}}\kappa$ is the dimensionless drag coefficient. Substituting $\mathbf{v}_{\text{L}} - \mathbf{v}_{\text{n}}$ from the last equation into the drag force $\mathbf{f}_{\text{d}}$ yields the mutual friction force density:

$$\begin{aligned} \mathbf{f}_{\text{MF}} &= \rho_{\text{s}}\kappa n_{\text{v}} \left[\frac{\mathcal{R}^2}{1 + \mathcal{R}^2} \hat{\omega} \times (\mathbf{v}_{\text{n}} - \mathbf{v}_{\text{s}}) \right. \\ &\left. + \frac{\mathcal{R}}{1 + \mathcal{R}^2} \hat{\omega} \times (\hat{\omega} \times (\mathbf{v}_{\text{n}} - \mathbf{v}_{\text{s}})) \right], \end{aligned} \quad (9)$$

where n_{v} is the vortex density. Assuming that the angular velocity of the normal crust is uniform, the internal torque is given by (A. Sedrakian et al. 1999):

$$\mathbf{N}_{\text{s}} = \int d^3r (\mathbf{r} \times \mathbf{f}_{\text{MF}}) = \mathbf{N}_{\text{s1}} + \mathbf{N}_{\text{s2}}, \quad (10)$$

where

$$\mathbf{N}_{\text{s1}} = \frac{\mathcal{R}}{1 + \mathcal{R}^2} I_{\text{s}} \left[\Omega_{\text{s}} \left(2 - \frac{\Omega_{\text{n}} \cdot \Omega_{\text{s}}}{\Omega_{\text{s}}^2} \right) \Omega_{\text{s}} - \Omega_{\text{s}} \Omega_{\text{n}} \right], \quad (11)$$

is the *dissipative* torque component, for which $\mathbf{N}_{\text{s1}} \cdot (\Omega_{\text{n}} - \Omega_{\text{s}}) < 0$, and

$$\mathbf{N}_{\text{s2}} = \frac{\mathcal{R}^2}{1 + \mathcal{R}^2} I_{\text{s}} (\Omega_{\text{s}} \times \Omega_{\text{n}}), \quad (12)$$

is the non-dissipative torque component. In general, the drag force is nonlinear, specifically, the drag parameter $\mathcal{R}$ depends on the relative velocity $\mathbf{v}_{\text{L}} - \mathbf{v}_{\text{n}}$ (e.g., T. Celora et al. 2020; X. Sheng et al. 2025). However, this nonlinearity should not affect our conclusions. Therefore, to simplify the analysis we adopt $\mathcal{R} = \text{const}$.

2.2. Euler's equations

As stated above, the two-component model adopted here treats the core and normal crust as a single deformed rigid body. It is, therefore, convenient to solve Euler's equations in the frame rotating with the normal crust. In this frame, the equations read:

$$\frac{d}{dt}\mathbf{L}_n + \boldsymbol{\Omega}_n \times \mathbf{L}_n = \mathbf{N}_{\text{ext}} + \mathbf{N}_s, \quad (13)$$

$$\frac{d}{dt}\mathbf{L}_s + \boldsymbol{\Omega}_n \times \mathbf{L}_s = -\mathbf{N}_s, \quad (14)$$

where $\mathbf{N}_{\text{ext}}$ is the external torque acting on the crust. For simplicity, we assume the moment of inertia tensor of the crustal superfluid to be isotropic, $\mathbf{I}_s = I_s \boldsymbol{\delta}$, where $\boldsymbol{\delta}$ is the unit tensor. It follows that $\mathbf{L}_s = I_s \boldsymbol{\Omega}_s$.

The moment of inertia of the coupled normal crust and core is, in general, anisotropic. Following A. Biryukov et al. (2025), we write its angular momentum in the principal-axis frame as $\mathbf{L}_n = I_0 \boldsymbol{\Omega}_n + I_0 \boldsymbol{\omega}$. Here, I_0 is the moment of inertia of the crust about the intermediate principal axis, and $\boldsymbol{\omega} = (\epsilon_1 \Omega_1, 0, \epsilon_3 \Omega_3)^T$, with $\epsilon_1 \leq 0$ and $\epsilon_3 \geq 0$.

With the above assumptions, Eqs. (13) and (14) yield:

$$\dot{\boldsymbol{\Omega}}_n + \dot{\boldsymbol{\omega}} = (\boldsymbol{\omega} \times \boldsymbol{\Omega}_n) + \frac{\mathbf{N}_{\text{ext}}}{I_0} + \frac{\mathbf{N}_{s1} + \mathbf{N}_{s2}}{I_0}, \quad (15)$$

and

$$\dot{\boldsymbol{\Omega}}_s = (\boldsymbol{\Omega}_s \times \boldsymbol{\Omega}_n) - \frac{\mathbf{N}_{s1} + \mathbf{N}_{s2}}{I_s}. \quad (16)$$

respectively.

3. APPROXIMATE ANALYTIC SOLUTIONS

In what follows, we consider, for simplicity, a biaxial star with ellipticity $\epsilon_3 = \epsilon$ and $\epsilon_1 \equiv 0$ without loss of generality. In the principal-axis frame, the vector $\boldsymbol{\omega}$ takes the form $\boldsymbol{\omega} = (0, 0, \omega)^T$, where

$$\omega = \epsilon \Omega_n \cos \theta \quad (17)$$

is the free-precession frequency in the absence of a superfluid component. Here, θ denotes the angle between $\boldsymbol{\Omega}_n$ and the symmetry axis (the wobble angle). The angular velocity of the normal crust can then be written in the principal frame as

$$\boldsymbol{\Omega}_n = \begin{pmatrix} \Omega_n \sin \theta \cos \varphi \\ \Omega_n \sin \theta \sin \varphi \\ \Omega_n \cos \theta \end{pmatrix}, \quad (18)$$

where φ is the corresponding azimuthal angle (see Figure 1). The angular velocity of the superfluid can be expressed similarly by replacing $\Omega_n, \theta, \varphi$ with $\Omega_s, \theta_s, \varphi_s$,

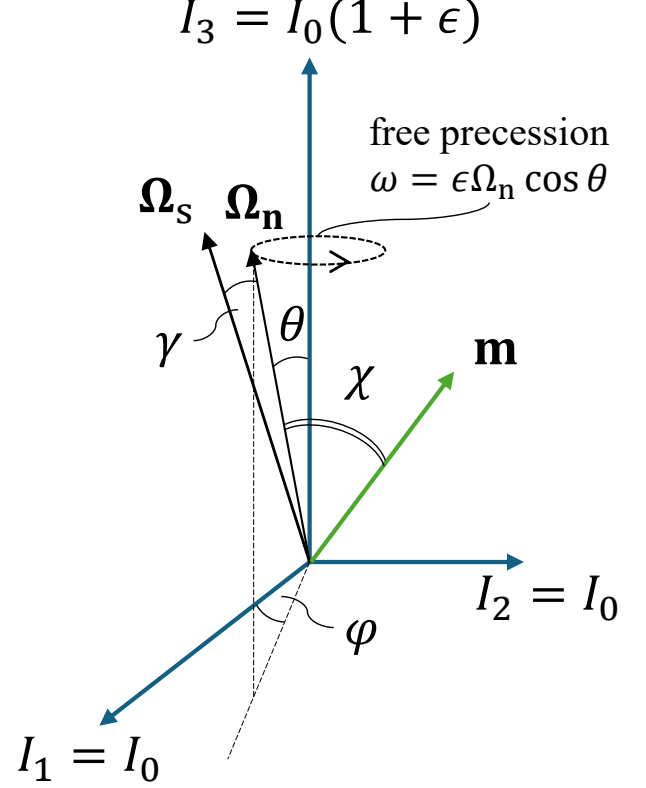


Figure 1. Reference frame aligned with the principal moments of inertia of the neutron-star normal crust, I_1 – I_2 – I_3 . In this work we consider mostly a biaxial star, such that $I_1 = I_2 = I_0$, where $I_0 = I_n + I_{co}$ is the combined moment of inertia of the normal crust and the core, whose rotation is assumed to be tightly coupled. The vector $\boldsymbol{\Omega}_n$ denotes the angular velocity of the normal crust, while $\boldsymbol{\Omega}_s$ is the angular velocity of the crustal superfluid component. The total effective deformation of the star is ϵ , so that a purely rigid body would precess about the symmetry axis (I_3) with angular velocity $\Omega_p = \epsilon \Omega_n \cos \theta$. The angles φ and θ are the spherical coordinates associated with this reference frame, with θ being the precession (wobbling) angle. The unit vector $\mathbf{m}$ is aligned with the magnetic axis and is fixed within this frame. Angle χ is the angle between the magnetic and spin axes.

respectively. The analytic solution is derived under the assumption that the ellipticity satisfies $|\epsilon| \ll 1$, implying that the free-precession period is much longer than the spin period $P_n = 2\pi/\Omega_n$. To first order in ϵ , the term $\dot{\boldsymbol{\omega}}$ on the LHS of Eq. (15) can therefore be neglected. Additionally, we consider that the external torque modifies the star's rotational evolution on a timescale much longer than the spin period

$$|\mathbf{N}_{\text{ext}}| \ll I_0 \Omega_n^2. \quad (19)$$

When the coupling of the superfluid and normal crust is sufficiently strong, the lag between the angular velocities of the superfluid and the normal crust, $\delta\mathbf{\Omega} \equiv \mathbf{\Omega}_s - \mathbf{\Omega}_n$, is also small:

$$|\delta\mathbf{\Omega}| = |\mathbf{\Omega}_s - \mathbf{\Omega}_n| \ll \Omega_n. \quad (20)$$

At the same time, if coupling is weak or moderate, then $\delta\mathbf{\Omega}$ will quickly drop to a small value, as shown below in Sect. 3.1. To leading order, the dissipative component of the internal torque can be approximated as:

$$\mathbf{N}_{s1} \approx I_s \Omega_n \frac{\mathcal{R}}{1 + \mathcal{R}^2} \left[\delta\mathbf{\Omega} + \left(\frac{\Omega_s}{\Omega_n} - 1 \right) \mathbf{\Omega}_n \right]. \quad (21)$$

Under the above assumptions, and subtracting Eq. (15) from (16), one obtains the evolution equation for $\delta\mathbf{\Omega}$:

$$\begin{aligned} \frac{d(\delta\mathbf{\Omega})}{dt} = & -(\boldsymbol{\omega} \times \mathbf{\Omega}_n) - \frac{\mathbf{N}_{\text{ext}}}{I_0} \\ & - (\mathbf{\Omega}_{\text{Sh}} \times \delta\mathbf{\Omega}) - \frac{\delta\mathbf{\Omega} + (\Omega_s/\Omega_n - 1)\mathbf{\Omega}_n}{\tau_d}, \end{aligned} \quad (22)$$

where

$$\tau_d = \frac{1 + \mathcal{R}^2}{\Omega_n \mathcal{R} (1 + \kappa_s)} \quad (23)$$

is the relaxation time due to the dissipative internal torque component $\mathbf{N}_{s1}$, and

$$\mathbf{\Omega}_{\text{Sh}} = -\mathbf{\Omega}_n \frac{1 - \kappa_s \mathcal{R}^2}{1 + \mathcal{R}^2}. \quad (24)$$

Hereafter $\kappa_s = I_0/I_s$. Note that in the limit $\mathcal{R} \rightarrow \infty$, $\tau_d \rightarrow \infty$ and $\mathbf{\Omega}_{\text{Sh}}$ reduces to wobble frequency derived by J. Shaham (1977). In the limit of zero drag, $\mathcal{R} \rightarrow 0$, the superfluid and crust are dynamically decoupled. The apparent rapid precession of the lag seen in the body frame is purely kinematic and results from the transformation to the rotating frame. In the axi-symmetric case, $\mathbf{\Omega}_s \parallel \mathbf{\Omega}_n$, Eq. (22) reduces to $\delta\dot{\Omega} = N_\Omega/I_0 + 2\delta\Omega/\tau_d$, where N_Ω is the spin-down torque, and it is seen that in this case $\tau_d/2$ is the damping time of the lag.

3.1. Quasi-steady rotational evolution

For a relaxation time much shorter than the spin-down time, $\tau_d \ll (n_\Omega \Omega_n)^{-1}$, the system relaxes to a quasi-steady state on a timescale $\sim \tau_d$, so that for $t \gg \tau_d$ Eq. (22) satisfies $\dot{\delta\mathbf{\Omega}} \approx 0$ (see Appendix B for details). A convenient decomposition of the lag is

$$\delta\mathbf{\Omega} = \delta\mathbf{\Omega}_\perp + \delta\mathbf{\Omega}_\parallel, \quad (25)$$

where

$$\begin{aligned} \delta\mathbf{\Omega}_\perp &= \Omega_n (\varphi_s - \varphi) \sin \theta \cdot \mathbf{e}_\varphi + \Omega_n (\theta_s - \theta) \cdot \mathbf{e}_\theta, \\ \delta\mathbf{\Omega}_\parallel &= (\Omega_s - \Omega_n) \mathbf{e}_\Omega, \end{aligned} \quad (26)$$

with the orthonormal basis vectors defined as: $\mathbf{e}_\Omega = \mathbf{\Omega}_n/\Omega_n$, $\mathbf{e}_\theta = (\cos \theta \cos \varphi, \cos \theta \sin \varphi, -\sin \theta)^T$ and $\mathbf{e}_\varphi = (-\sin \varphi, \cos \varphi, 0)^T$. To linear order $|\delta\mathbf{\Omega}_\perp|/\Omega_n$ is the angle γ between $\mathbf{\Omega}_n$ and $\mathbf{\Omega}_s$. Solving Eq. (22) for $\delta\dot{\mathbf{\Omega}} = 0$ gives the equilibrium components of $\delta\mathbf{\Omega}$:

$$\begin{aligned} |\delta\mathbf{\Omega}_{\perp\text{eq}}|^2 &\approx \Omega_n^2 \gamma_{\text{eq}}^2 = \\ &= \Omega_n^2 [(\epsilon \sin \theta \cos \theta + n_\varphi)^2 + n_\theta^2] \frac{1 + \mathcal{R}^2}{1 + \kappa_s^2 \mathcal{R}^2}, \end{aligned} \quad (27)$$

and

$$\delta\Omega_{\parallel\text{eq}} = -\Omega_n \frac{n_\Omega(1 + \mathcal{R}^2)}{2\mathcal{R}(1 + \kappa_s)}. \quad (28)$$

Here n_i are the dimensionless projections of the external torque,

$$n_i = \mathbf{N}_{\text{ext}} \cdot \mathbf{e}_i / I_0 \Omega_n^2, \quad i = \Omega, \theta, \varphi. \quad (29)$$

Note that since $\epsilon, n_i \ll \kappa_s \ll 1$ for a realistic neutron star, $\gamma_{\text{eq}} \ll 1$ for any value of the drag parameter $\mathcal{R}$. The condition $\delta\Omega_{\parallel\text{eq}} \ll \Omega_n$ is satisfied provided that $\Omega_n \tau_d \ll n_\Omega^{-1}$. In case of free precession, $n_\Omega = 0$, the equilibrium value of $\delta\Omega_{\parallel}$ must be determined to second order. Omitting the details, one finds that

$$\delta\Omega_{\parallel\text{eq}} = -\Omega_n \frac{n_\Omega(1 + \mathcal{R}^2)}{2\mathcal{R}(1 + \kappa_s)} - \Omega_n \frac{\gamma_{\text{eq}}^2}{2(1 + \kappa_s)}. \quad (30)$$

For realistic stars $n_\Omega \gg \gamma_{\text{eq}}^2$, so that a linear expansion provides a good approximation for the lag. The second-order term is included here only to enable comparison between the analytical and numerical solutions in the case of free precession.

The relaxation toward quasi-steady evolution in the case of free precession is illustrated in Fig. 2, which shows numerical solutions of Eqs. (15) and (16) for different parameter values. It indicates that the characteristic time to reach the quasi-steady state is τ_d , as naively expected. A good agreement is seen between the analytic values of γ_{eq} and $\delta\Omega_{\text{eq}}$ given in Eqs. (27) and (30) (the horizontal dashed lines in Fig. 2), and the numerical solutions.

Upon substituting $\mathbf{\Omega}_s = \mathbf{\Omega}_n + \delta\mathbf{\Omega}_{\text{eq}}$ into Eq. (15), one obtains the secular evolution equations for the wobbling angle θ and the azimuthal angle φ :

$$\dot{\theta} = \Omega_n \frac{n_\theta - (\epsilon \sin \theta \cos \theta + n_\varphi) \kappa_s \mathcal{R}}{1 + \kappa_s^2 \mathcal{R}^2}, \quad (31)$$

$$\dot{\varphi} = \Omega_n \frac{\epsilon \sin \theta \cos \theta + n_\varphi + n_\theta \kappa_s \mathcal{R}}{\sin \theta (1 + \kappa_s^2 \mathcal{R}^2)}, \quad (32)$$

restricted to the range $\mathcal{R} \ll (1 + \kappa_s)/n_\Omega$. As will be clarified below, Eq. (31) governs the decay of slow precession modes, owing to the coupling of the crustal superfluid

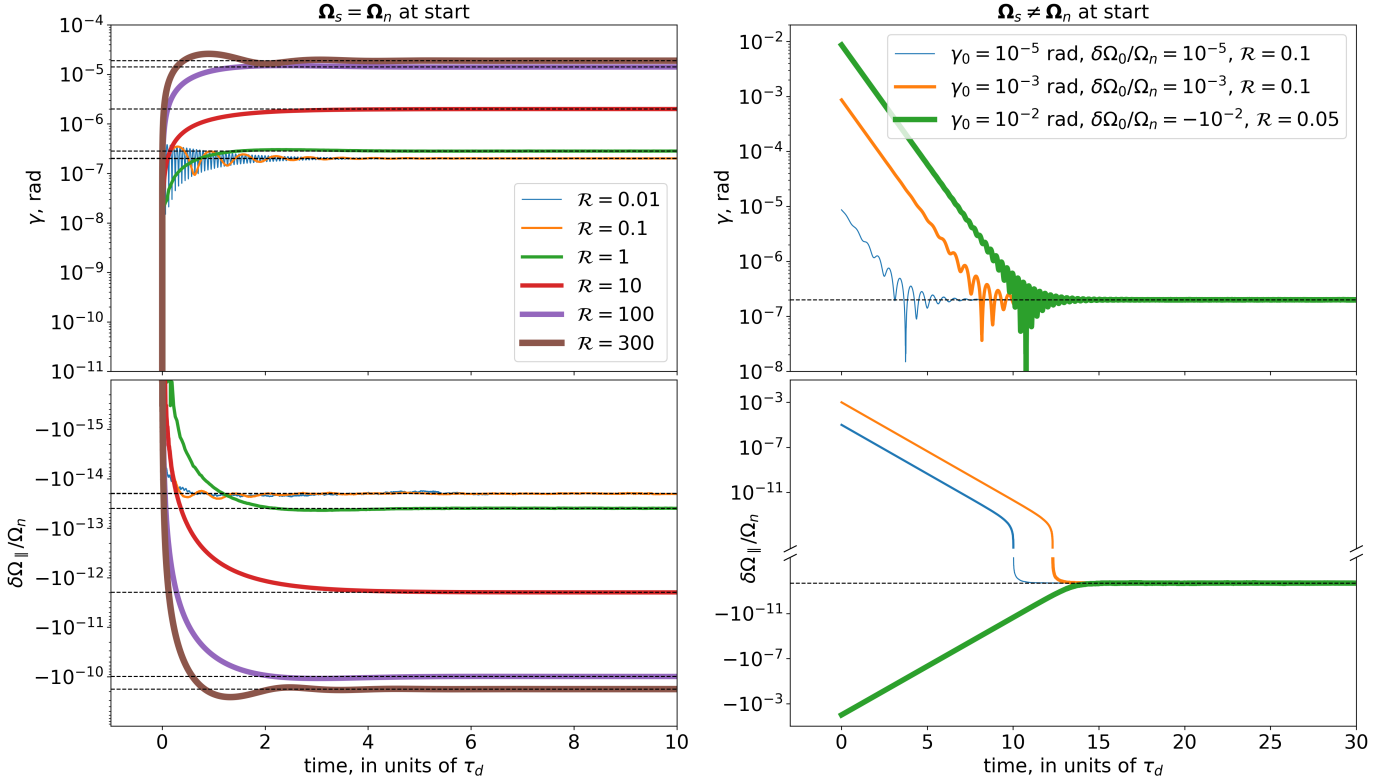


Figure 2. Evolution of γ and $\delta\Omega_{||}/\Omega_n = \Omega_s/\Omega_n - 1$ for free precession ($\mathbf{N}_{\text{ext}} = 0$), with $P_{n,0} = 1$ s, $\epsilon = 4 \times 10^{-7}$, $\kappa_s = 0.01$, and initial tilt $\theta_0 = 45^\circ$. The left panels show solutions for different values of the drag parameter $\mathcal{R}$, assuming the initial condition $\Omega_n = \Omega_s$. The right panels show solutions for different initial values of γ and $\delta\Omega_{||} = \Omega_s - \Omega_n$, as indicated. Solid lines represent numerical solutions of Eqs. (15)–(16), while horizontal dashed lines denote the analytic equilibrium values given by Eqs. (27) and (30).

and the normal crust. The decay of the fast (Shaham) mode occurs during the rapid transient phase, over time $t \lesssim \tau_d$ (see Appendix B). The spin-down equation for the normal crust (and the coupled core) is obtained, to linear order, by projecting Eq. (15) onto $\mathbf{e}_\Omega$:

$$\begin{aligned} \dot{\Omega}_n &= \Omega_n^2 n_\Omega + \frac{2\kappa_s \mathcal{R}}{1 + \mathcal{R}^2} \Omega_n \delta\Omega_{||\text{eq}} \\ &= \frac{\Omega_n^2 n_\Omega}{1 + \kappa_s} = \frac{\mathbf{N}_{\text{ext}} \cdot \mathbf{e}_\Omega}{I_0 + I_s}. \end{aligned} \quad (33)$$

We see that the spin-down rate of the crust coincides with that of a rigid body, as required by angular momentum conservation. It is instructive to first consider the limit of free precession, as it captures, to a good approximation, the evolution of the system in the general case when $n_i \ll \epsilon$, as in the case of Her X-1. In this limit, Eqs. (31) and (32) reduce to

$$\dot{\varphi} = \frac{\epsilon \Omega_n \cos \theta}{1 + \kappa_s^2 \mathcal{R}^2}, \quad (34)$$

and

$$\dot{\theta} = -\frac{\epsilon \Omega_n \kappa_s \mathcal{R}}{1 + \kappa_s^2 \mathcal{R}^2} \cos \theta \sin \theta. \quad (35)$$

The solution of the last equation can be written as:

$$\tan(\theta) = \tan(\theta_0) \cdot \exp\left(-\frac{t}{\tau_p}\right), \quad (36)$$

where

$$\tau_p = \frac{1 + \kappa_s^2 \mathcal{R}^2}{\Omega_n \epsilon \kappa_s \mathcal{R}} \approx \frac{(1 + \kappa_s^2 \mathcal{R}^2)}{\epsilon \kappa_s (1 + \mathcal{R}^2)} \tau_d > \frac{\kappa_s \tau_d}{\epsilon}. \quad (37)$$

As seen, for $\epsilon \ll \kappa_s$ we have $\tau_p \gg \tau_d$. Note that the evolution of θ depends on the sign of ϵ (see Fig. 3). For $\epsilon > 0$ (an oblate star), $\theta \rightarrow 0$ as $t \gg \tau_p$, and the amplitude of the precession decays. For $\epsilon < 0$ (a prolate star), $\theta \rightarrow \pi/2$ as $t \gg \tau_p$, while $\dot{\varphi} \rightarrow 0$, i.e. the precession period diverges. The evolution of θ reflects the tendency of the system to minimize its rotational energy at fixed angular momentum. In either case, the slow free-precession mode damps on a timescale $\sim \tau_p$. In other words, the slow precession persists for $\gtrsim (\kappa_s \mathcal{R})^{-1}$ cycles independently on ϵ .

4. APPLICATION TO HERCULES X-1

The origin of the ~ 35 -day modulation observed in the X-ray binary Her X-1 (H. Tananbaum et al. 1972;

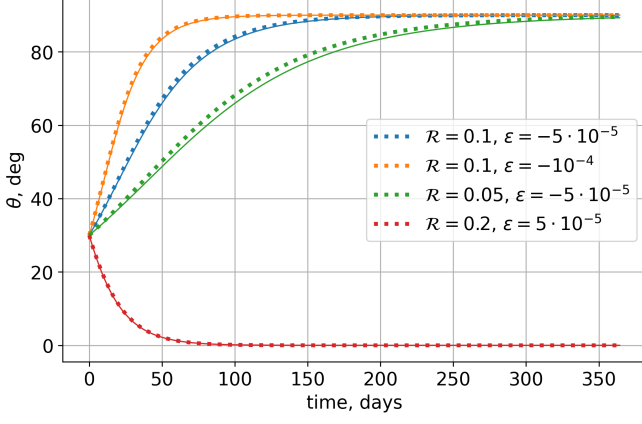


Figure 3. The evolution of the wobbling angle θ in the limit of free precession, for $\kappa_s = 10^{-2}$, and for different values of the drag parameter $\mathcal{R}$ and ellipticity ϵ , as indicated. The initial crustal spin period and tilt are $P_{n,0} = 1$ s and $\theta_0 = 30^\circ$, respectively. The dotted lines show the exact numerical solutions of Eqs. (15) and (16), while the solid lines show the corresponding analytical solutions given by Eq. (36). Good agreement is obtained.

R. Giacconi et al. 1973) has been debated for several decades. A widely accepted interpretation attributes the cycle to the forced precession of a tilted and warped accretion disk, which periodically obscures the central X-ray source and produces the characteristic sequence of main-on, short-on and off states in the X-ray light curve (J. I. Katz 1973; J. A. Petterson 1975; S. Schandl & F. Meyer 1994; R. A. M. J. Wijers & J. E. Pringle 1999). In this picture, tidal torques from the companion star together with radiation-driven warping lead to a precessing disk whose varying orientation modulates the observed flux. An alternative interpretation invokes free precession of the neutron star, motivated in part by the systematic evolution of the X-ray pulse profiles over the 35-day cycle, which may reflect a changing orientation of the emission beam relative to the observer (J. Truemper et al. 1986; N. I. Shakura et al. 1998; K. Postnov et al. 2013; D. Kolesnikov et al. 2022). Recent X-ray polarimetric observations with the Imaging X-ray Polarimetry Explorer have provided new evidence supporting this possibility. Phase-resolved polarization measurements across the superorbital cycle reveal systematic variations in the polarization angle and degree that are consistent with changes in the viewing geometry of the magnetic poles expected if the neutron star itself undergoes free precession (J. Heyl et al. 2024; Q. C. Zhao et al. 2024). These observations suggest that the ~ 35 -day clock may be set by the free precession of the neutron-star crust, implying a small but finite asymmetry in the star’s moment of inertia (of order a few parts per 10^7) required to sustain such motion. Hybrid scenar-

ios have also been proposed in which both the neutron star and the accretion disk precess, forming a system of coupled clocks in which the disk primarily controls the X-ray turn-ons while neutron-star precession modulates the pulse-profile morphology (e.g., K. A. Postnov et al. 2000; R. Staubert et al. 2009). The relative roles of these mechanisms remain an active topic of investigation.

In what follows, we consider the implications of neutron star precession for the superfluid dynamics. Under the assumption of tight core-crust coupling, the net external torque acts on the star on timescales much longer than the observed 35-d cycle in Her X-1, which in the free-precession interpretation is driven by a stellar deformation with ellipticity $\epsilon \approx 4 \times 10^{-7}$ (e.g., J. Heyl et al. 2024). More precisely,

$$N_{\text{ext,HX1}} \ll I_0 \Omega_n^2 |\epsilon|, \quad (38)$$

where $N_{\text{ext,HX1}}$ is the magnitude of the net external torque, which can be decomposed into several components,

$$\mathbf{N}_{\text{ext,HX1}} = \mathbf{N}_{\text{acc}} + \mathbf{N}_{\text{mag}} + \mathbf{N}_{\text{psr}} + \mathbf{N}_{\text{warp}} + \mathbf{N}_{\text{prec}}. \quad (39)$$

From left to right, the terms on the right-hand side represent the accretion spin-up torque, magnetospheric spin-down torque, pulsar losses, and the back-reaction associated with the warping and precession of the accretion disc, respectively. The latter two are due to the interaction of the inclined disc with the NS magnetosphere (D. Lai et al. 2011; D. Lai 1999). The details are provided in Appendix C.

It is worth noting that $\mathbf{N}_{\text{acc}}$, $\mathbf{N}_{\text{warp}}$, and $\mathbf{N}_{\text{prec}}$ rotate rapidly about $\boldsymbol{\Omega}_n$ in the principal frame of the star, as they are linked to the accretion disc. Thus, their contribution effectively averages out over the spin period. The corresponding spin-averaged torque can be written as the sum of two components, parallel and orthogonal to the crust’s spin axis, respectively:

$$\langle \mathbf{N}_{\text{ext,HZ1}} \rangle_{\text{spin}} = \mathbf{N}_{\parallel} + \mathbf{N}_{\perp}. \quad (40)$$

The first component is responsible for the neutron star spin-down:

$$\mathbf{N}_{\parallel} = \left[N_{\text{psr},0} (1 + \sin^2 \chi) + N_{\text{mag}} + N_{\text{acc},0} \cos \alpha \left(1 + \frac{4\sqrt{2}}{3} \sin^2 \alpha \cos^2 \chi \right) \right] \mathbf{e}_{\Omega}, \quad (41)$$

while the second one is capable of changing the orientation of $\boldsymbol{\Omega}_n$ with respect the principal frame:

$$\mathbf{N}_{\perp} = N_{\chi} ((\mathbf{m} \times \mathbf{e}_{\Omega}) \times \mathbf{e}_{\Omega}) + N_{\text{rad}} (\mathbf{m} \times \mathbf{e}_{\Omega}), \quad (42)$$

Here $\mathbf{m}$ is the unit vector aligned with the magnetic axis. It is assumed to be fixed in the principal frame, and misaligned with respect to the deformation axis. The torque

$$N_\chi = (N_{\text{psr},0} + \eta A N_{\text{acc},0} \sin^2 \alpha \cos \alpha) \cos \chi \quad (43)$$

is responsible for the star magnetic obliquity evolution, while

$$N_{\text{rad}} = \frac{N_{\text{psr},0}}{3} \left(\frac{r_{\text{LC}}}{r_{\text{NS}}} \right) \cos \chi \quad (44)$$

is the pulsar “anomalous” torque (e.g. V. S. Beskin & A. A. Zheltoukhov 2014). Here $\chi = \cos^{-1}(\mathbf{m} \cdot \mathbf{e}_\Omega)$ is the magnetic angle, α is the angle between the accretion-disk axis and $\mathbf{\Omega}_n$, $r_{\text{LC}} = c/\Omega_n$ is the light-cylinder radius, and r_{NS} is the neutron-star radius. The dimensionless parameters $\eta = 0.1$ and

$$A = \left[1 - \frac{\eta}{2} (\sin^2 \chi \sin^2 \alpha + 2 \cos^2 \chi \cos^2 \alpha) \right]^{-1}, \quad (45)$$

describe the spin-modulated accretion torque.

The orthogonal component, $\mathbf{N}_\perp$, is the only one capable of influencing the time evolution of the wobbling angle, θ . Indeed

$$n_{\theta,\varphi} \equiv \frac{N_{\theta,\varphi}}{I_0 \Omega_n^2} = \frac{\mathbf{N}_\perp \cdot \mathbf{e}_{\theta,\varphi}}{I_0 \Omega_n^2}, \quad (46)$$

while $\mathbf{N}_\perp \cdot \mathbf{e}_\Omega \equiv 0$, and $n_\Omega \equiv N_\parallel / I_0 \Omega_n^2$. For the estimates below, we adopt parameters consistent with those inferred for the neutron star in Her X-1. In particular, we take a spin period $P_{\text{n,HX1}} = 1.2$ s, a magnetic moment $\mu_{\text{HX1}} = 3 \times 10^{30}$ G cm³ (R. Staubert et al. 2014), and a magnetospheric radius $r_m = 10^3$ km. The latter is a conservative order-of-magnitude estimate that remains smaller than the corotation radius of the system, $r_c \approx 2 \times 10^3$ km (for a $1.5 M_\odot$ neutron star). Since $r_m < r_c$, the system is expected to remain in the accretion regime. However, r_m for Her X-1 is not well constrained observationally; estimates in the literature span a broad range, from ~ 100 to 4×10^3 km (M. B. Averintsev et al. 1992; D. M. Scott et al. 2000; G. Ramsay et al. 2002). Within the adopted torque model (Appendix C), $r_m = 10^3$ km corresponds to an accretion rate $\dot{M}_{\text{HX1}} = 5 \cdot 10^{-8} M_\odot \text{ yr}^{-1}$. This value is about an order of magnitude larger than observational estimates (B. S. Boroson et al. 2007; P. Kosec et al. 2020); however, since the spin-down timescale of the star is much longer than the precession period, the resulting torque estimates remain representative for our purposes. Namely,

$$N_{\text{acc},0} = (4.7 \times 10^{35} \text{ dyn cm}) \left(\frac{\mu}{\mu_{\text{HX1}}} \right)^{2/7} \left(\frac{\dot{M}}{\dot{M}_{\text{HX1}}} \right)^{6/7}, \quad (47)$$

$$N_{\text{mag}} = (-4.1 \times 10^{35} \text{ dyn cm}) \left(\frac{\mu}{\mu_{\text{HX1}}} \right)^2 \left(\frac{P}{P_{\text{n,HX1}}} \right)^{-2}, \quad (48)$$

and

$$N_{\text{psr},0} = (-4.8 \times 10^{31} \text{ dyn cm}) \left(\frac{\mu}{\mu_{\text{HX1}}} \right)^2 \left(\frac{P}{P_{\text{n,HX1}}} \right)^{-3}. \quad (49)$$

Notice that $(r_{\text{LC}}/r_{\text{NS}})N_{\text{psr},0}/3 \approx -10^{35}$ dyn cm for $r_{\text{NS}} = 10$ km, implying that the pulsar torque can contribute equally to the precessional evolution of the star.

For the order-of-magnitude estimates below, we neglect geometric factors and adopt $N_{\text{ext,HX1}} \sim N_\perp \sim N_\parallel \sim 10^{35}$ dyn cm, given that $I_0 = 10^{45}$ g cm², $P_n = 1.2$ s, and $|\epsilon| = 4 \cdot 10^{-7}$ (J. Heyl et al. 2024). Thus one gets

$$\frac{N_{\text{ext,HX1}}}{I_0 \Omega_n^2 |\epsilon|} \approx 10^{-5}, \quad (50)$$

so that the external torque is negligible compared to the deformation torque. On the other hand, as total $\dot{\theta} \propto n_\theta - (\epsilon \sin \theta \cos \theta + n_\varphi) \kappa_s \mathcal{R}$, then for

$$\mathcal{R} \gtrsim 0.002 N_{\perp,35} |\epsilon|_{-7}^{-1} \kappa_{s,-2}^{-1} I_{0,45}^{-1} P_{\text{n,s}}^2, \quad (51)$$

the internal torque is also substantially stronger than the external one. Here, $N_{\perp,35} = N_\perp / (10^{35} \text{ dyn cm})$, $|\epsilon|_{-7} = |\epsilon| / 10^{-7}$, $I_{0,45} = I_0 / (10^{45} \text{ g cm}^2)$ and $\kappa_{s,-2} = \kappa_s / 10^{-2}$.

In the case of weak pinning (Eq. 51 is not fulfilled), and if $\epsilon \cos \theta$ and n_θ have the same signs, the latter can be large enough to compensate, in principle, the dissipation of free precession due to crust-superfluid interaction.

This might be possible as $\mathbf{N}_{\text{ext}}$ consists of components which affect the orientation of the rotation axis with respect to the principal axes of the crust. One is the last term in $\mathbf{N}_\perp$ related to pulsar emission, and the other is the term proportional to η which arises from the accretion torque modulation with the spin phase (see Eq. 43).

However, a detailed analysis of the specific conditions that must be met for such a balance is beyond the scope of this work.

In the opposite case, if external torque does not compensate the dissipation, the neutron star precesses essentially freely, and the evolution of θ is controlled by the superfluid component. Even for $\mathcal{R} = 0.01$, the precession of Her X-1 neutron star damps on a timescale of at least $(\kappa_s \mathcal{R})^{-1} \approx 10^4$ cycles, corresponding to $\approx 10^3$ years. This is much longer than the ~ 50 -year observational time span of the system.

To demonstrate this, we solved Eqs. (15) and (16) including the external torque (40), adopting $\chi_0 = 5^\circ$ (with the magnetic axis tilted relative to the deformation axis

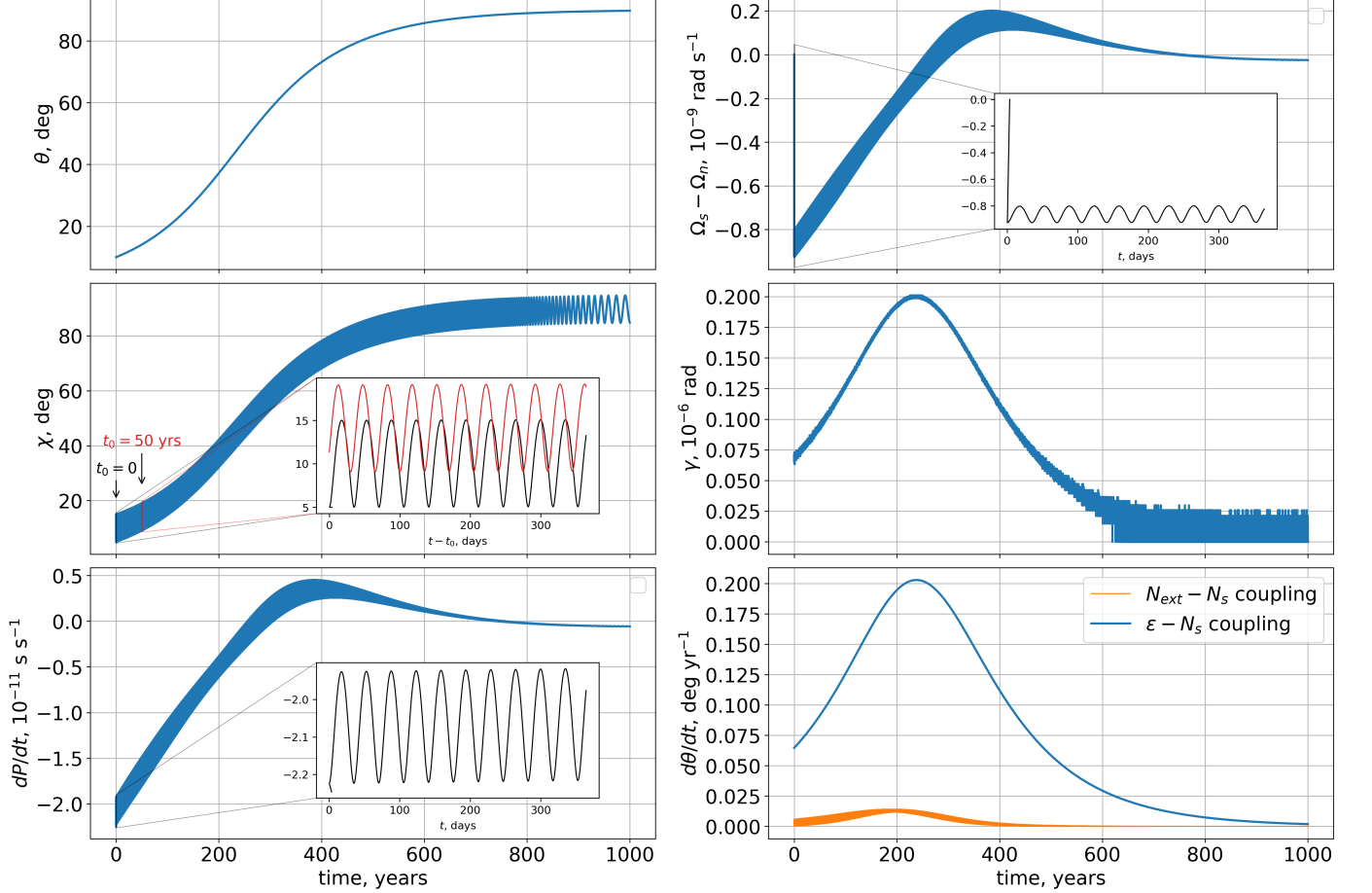


Figure 4. Possible evolutionary track of the neutron star in Her X-1, taking into account the internal torque exerted by the crustal superfluid. The plots show the numerical solution of Eqs. (15)–(16) including the external torque given by Eq. (40). The star is assumed to be prolate with ellipticity $\epsilon = -4 \times 10^{-7}$ and superfluid drag coefficient $\mathcal{R} = \kappa_s = 0.01$; the accretion rate is $\dot{M} = 5 \times 10^{-8} M_\odot \text{ yr}^{-1}$. In the left column, the evolution of the wobbling angle θ , magnetic angle χ , and spin-down rate $\dot{P}$ are shown (from top to bottom). In the right column, the angular-frequency lag $\delta\Omega_{||} = \Omega_s - \Omega_n$ and the tilt γ are shown in the top and middle panels, respectively. The bottom-right panel shows the contributions to $\dot{\theta}$ described by Eq. (31): the term $\propto -\epsilon \sin \theta \cos \theta \kappa_s \mathcal{R}$ is shown by the blue curve, while the term $\propto n_\theta - n_\varphi \kappa_s \mathcal{R}$ is shown by the orange curve. The evolution of the wobbling angle is therefore predominantly driven by the superfluid torque coupled to the stellar deformation.

by an angle $\theta_m = 10^\circ$) and $\alpha_0 = 25^\circ$. These values are consistent with the parameters provided for the Her X-1 system (see e.g. J. Heyl et al. 2024). The amplitude of the magnetic angle variations during the 35-day cycle, estimated from X-ray polarization, is only a few degrees, implying that the crust spin axis and the magnetic axis are nearly aligned with the symmetry axis. We also assumed a prolate star ($\epsilon = -4 \times 10^{-7}$) with small loss of generality. On the other hand, negative ϵ could be justified while the internal toroidal magnetic field tends to distort a star into a prolate shape (C. Cutler 2002).

The spin-disc tilt α was evolved self-consistently as

$$I_0 \Omega_n \dot{\alpha} = -N_{\text{acc},0} \sin \alpha \left(1 - \frac{4\sqrt{2}}{3} \cos^2 \alpha \cos^2 \chi \right). \quad (52)$$

The superfluid parameters were set to $\mathcal{R} = \kappa_s = 0.01$, and the accretion torque modulation parameter to $\eta = 0.99$. The system has evolved over 1000 years.

Figure 4 shows the results. The left column displays θ , χ , and $\dot{P}$ (top to bottom), while the right column shows the crust–superfluid lag $\delta\Omega$ and tilt γ . The bottom-right panel decomposes $\dot{\theta}$ into the internal contribution $\propto -\epsilon \sin \theta \cos \theta \kappa_s \mathcal{R}$ (blue) and the external torque term $\propto n_\theta - n_\varphi \kappa_s \mathcal{R}$ (orange). For these parameters, the external torque has a negligible impact on θ , and the system follows a free-precession-like evolution. Since $\epsilon < 0$, the wobbling angle grows secularly on $\sim 10^2$ -yr timescales, consistent with Eq. (31), while χ exhibits nearly constant modulation amplitude with a slow drift in its cycle-averaged value, $\langle \chi \rangle$, by $\sim 5^\circ$ over 50 years. Spin-down modulations (Fig. 4, lower left) are dominated by the

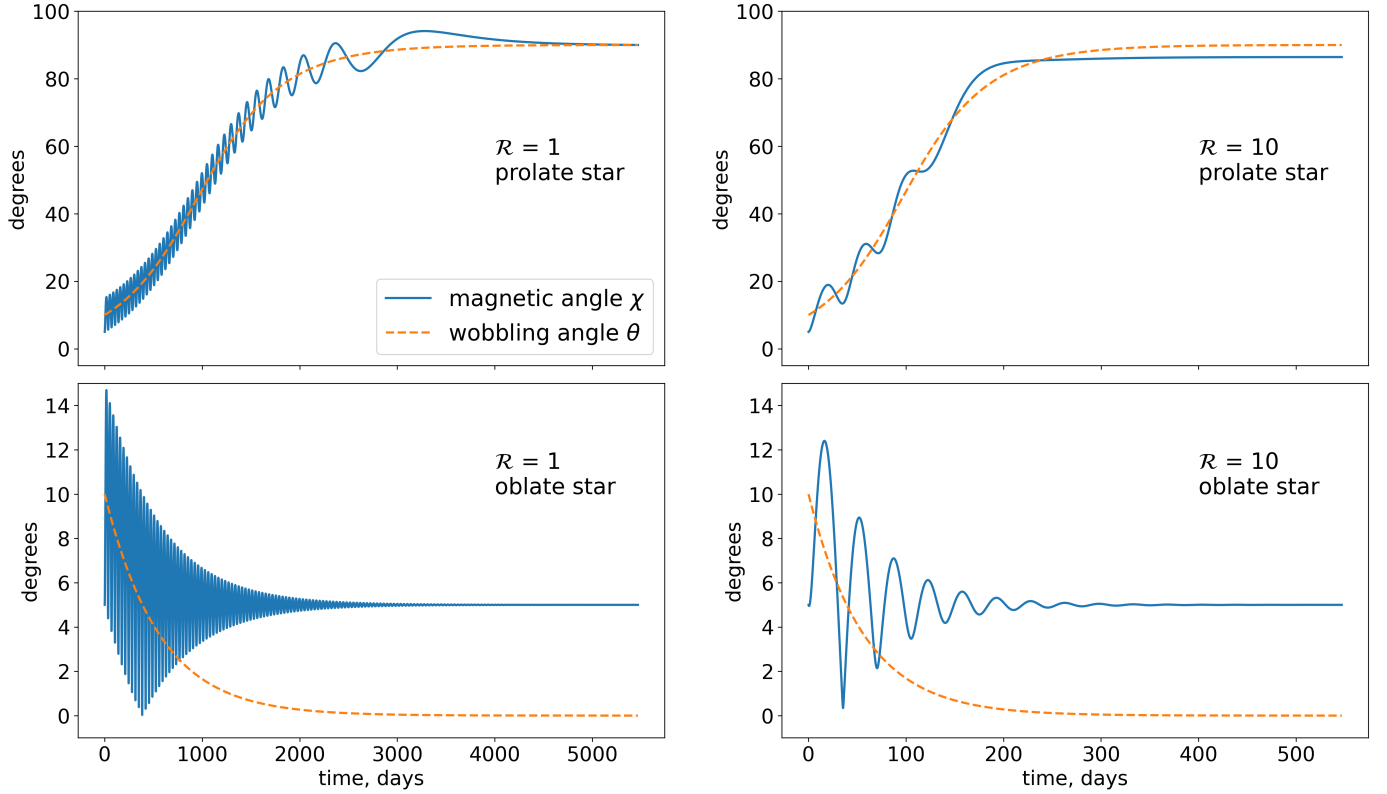


Figure 5. Modeled evolution of the magnetic angle χ and the wobbling angle θ of Her X-1 for $\mathcal{R} = 1$ (left panels) and $\mathcal{R} = 10$ (right panels), for prolate (top) and oblate (bottom) stars. All the other parameters are the same as in Figure 4.

warping torque term in Eq. (41) and exceed observational estimates by roughly an order of magnitude (cf. Fermi GBM data³; C. Meegan et al. 2009; D. Kolesnikov et al. 2022), reflecting that the parameters are not fully tuned to observations. Nevertheless, even this enhanced accretion torque does not alter the precessional evolution, since the internal contribution to $\dot{\theta}$ remains dominant.

For completeness, we also computed the rotational evolution of Her X-1 using the same parameters but with stronger superfluid drag, $\mathcal{R} = 1$ and 10. The results are shown in the left and right columns of Fig. 5, respectively. We further considered both prolate ($\varepsilon = -4 \times 10^{-7}$) and oblate ($\varepsilon = +4 \times 10^{-7}$) configurations to illustrate the effect of precession damping predicted by Eq. (35). The results are fully consistent with the analytical expectations: increasing drag accelerates the evolution of the precession, driving the wobbling angle toward either 0° or 90° , depending on the sign of ε . In the calculations above, the Her X-1 neutron star was assumed to be biaxial, which is not necessarily the case. For example, D. Kolesnikov et al. (2022) argued that the observed 35-d spin variations are consistent

with triaxial precession of the star. Analyzing pulsar spin-down history, they estimate the system parameters as $\theta = 50^\circ$, with the magnetic axis oriented along $(\varphi_m, \theta_m) = (90^\circ, 30^\circ)$. Adopting these parameters as an initial rotational state, along with $\varepsilon_3 = -\varepsilon_1 = 4 \times 10^{-7}$, and keeping all other parameters the same as used for the biaxial case, we performed simulations of the spin evolution for a triaxial configuration. The results are shown in Fig. 6.

The rotational fate of the star is qualitatively similar to that in the biaxial case. The spin axis aligns with one of the principal axes on the timescale set by the dissipative component of the internal torque. During the first ~ 100 years of evolution, the star precesses around the minor inertial axis with large-amplitude variations in θ and χ . It then transitions to a different precessional state on an orbit around the principal axis with the maximal moment of inertia. This behavior is fully consistent with expectations for triaxially deformed neutron stars with dissipative torques; see A. Biryukov et al. (2025) for a detailed theoretical discussion.

Our final conclusion is that for sufficiently weak drag, $\mathcal{R} \lesssim 0.01$, the neutron star in Her X-1 can sustain precession for at least hundreds of years, consistent with observational constraints. Higher values of the drag parameter $\mathcal{R}$ would be inconsistent with the observations.

³ <https://gammaray.nsstc.nasa.gov/gbm/science/pulsars/lightcurves/herx1.html>

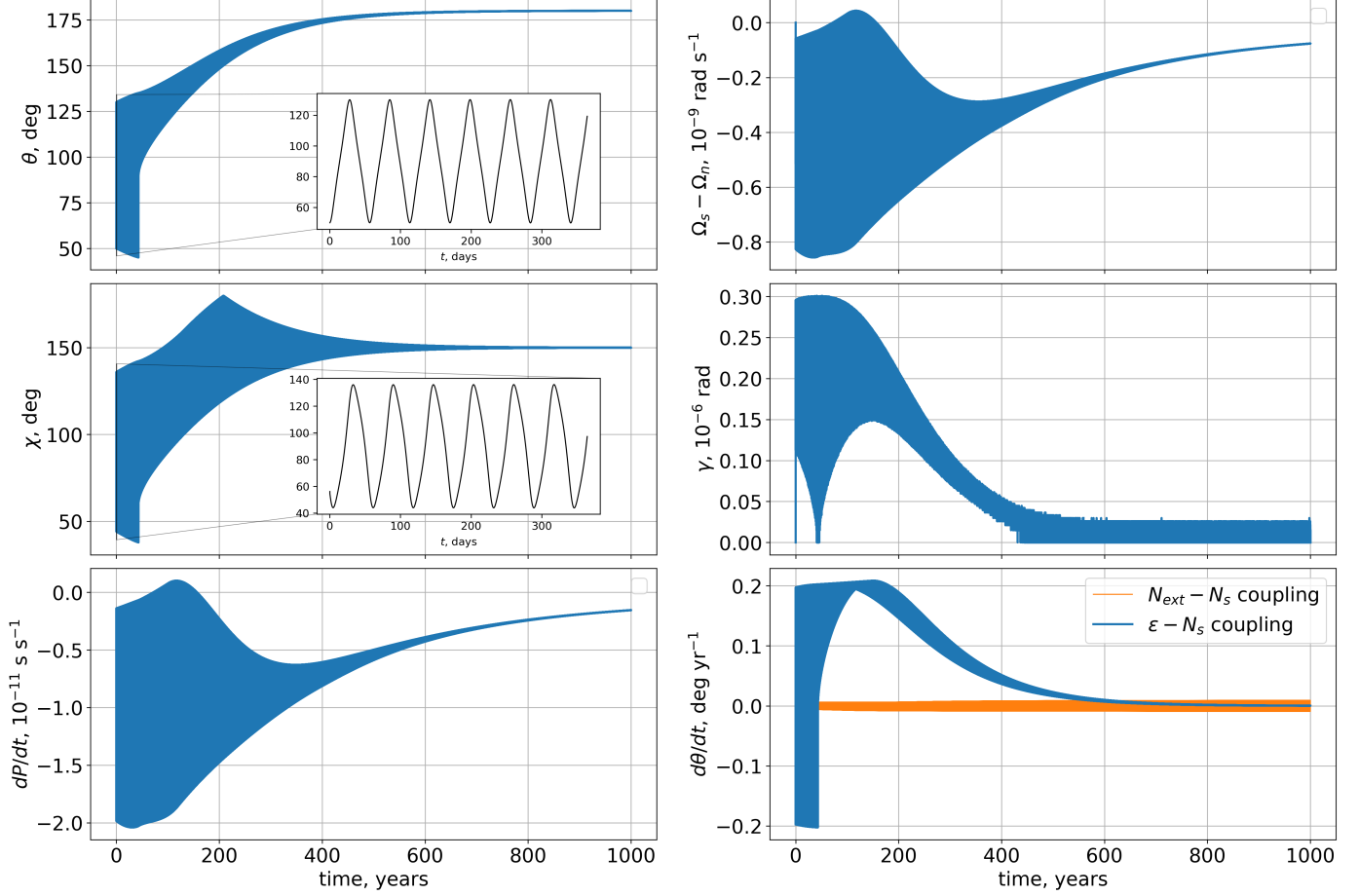


Figure 6. Possible evolution of the neutron star in Her X-1 assuming a triaxial deformation $\epsilon_3 = -\epsilon_1 = 4 \times 10^{-7}$ and initial orientations of $\mathbf{\Omega}_n$ and $\mathbf{m}$ as estimated by D. Kolesnikov et al. (2022). In particular, $\theta_0 = 50^\circ$, and $\mathbf{m}$ points in the direction $(\varphi_m, \theta_m) = (90^\circ, 30^\circ)$. The terms contributing to $\dot{\theta}$ in the bottom right plot were calculated using the Eq.(31) assuming $\epsilon = \sqrt{\epsilon_1^2 + \epsilon_3^2}$. Although this equation is formally derived for the biaxial case and therefore does not strictly apply to a triaxial star, it still provides a reasonable theoretical estimate of $\dot{\theta}$. All other parameters are the same as in Figure 4. The stellar evolution is qualitatively similar to the biaxial case; during the first ~ 100 years the spin axis precesses around the minor inertia axis (ϵ_1), with large-amplitude variations of θ and χ . Subsequently, it transitions to precession about the major inertia axis ($\theta \rightarrow 180^\circ$) on a timescale determined by the dissipative internal torque.

5. DISCUSSION

The very low effective drag, $\mathcal{R} < 10^{-2}$, implied by our analysis of the neutron star free precession in Her X-1, carries important consequences for the standard picture of superfluid dynamics and glitch phenomenology. In conventional glitch models, rapid spin-up events are attributed to the sudden unpinning and redistribution of angular momentum stored in a strongly pinned superfluid component, with subsequent coupling to the crust mediated by finite mutual friction and vortex creep. Unpinning occurs once the lag developed during the spin-down of the crust approaches the critical lag,

$$\Delta\Omega_{\text{crit}} \simeq \frac{f_p}{\rho_s \kappa r_{\text{NS}}} \sim 0.1 - 1 \quad \text{rad s}^{-1}, \quad (53)$$

where f_p is the pinning force per unit length, ρ_s the superfluid density, and κ the quantum of circulation. This

is larger by several orders of magnitude than the maximum lag developed during the quasi-steady precession phase (cf. Eq. 27):

$$|\mathbf{\Omega}_s - \mathbf{\Omega}_n| \lesssim \Omega_n \epsilon \approx 10^{-6} \quad \text{rad s}^{-1}. \quad (54)$$

Consequently, the relative motion between the superfluid and the normal component induced by precession is far too small to trigger vortex unpinning. In fact, the projected fractional lag obtained from Eq. (28) for the net torque inferred in Her X-1,

$$\frac{|\delta\Omega_{||\text{eq}}|}{\Omega_n} \lesssim 10^{-9}, \quad (55)$$

is several orders of magnitude smaller than the fractional spin jumps observed in glitches in many pulsars. This suggests that the dynamical state realized during long-lived precession is incompatible with the buildup of the

large differential rotation required for glitch triggering. If the superfluid responsible for storing angular momentum in glitching pulsars behaves similarly in accreting systems such as Her X-1, the results imply that either the regions involved in glitches are largely decoupled from the precessing component, or that the effective vortex drag and pinning conditions depend sensitively on the dynamical state of the star.

Alternatively, the 35-day cycle may arise from a different mechanism, and the IXPE observations may admit an alternative interpretation. A low-mass third body orbiting the system has been proposed to explain the light curve (K. Brecher & I. Wasserman 1974; T. Mazeh & J. Shaham 1977). However, this model predicts eccentricity of a few percents for the main binary orbit, which has been ruled out by accurate pulse-timing observations (J. E. Deeter et al. 1981). It is also difficult to see how such a model could account for the IXPE data.

C. L. Wolff & Y. Kondo (1978) proposed an alternative scenario in which the periodicities in the light curve of Her X-1 are associated with oscillatory modes (g-modes) of the companion star, HZ Her. However, this model predicts additional periodicities on a timescale of ~ 1 day, which are not observed. Moreover, it does not account for the periodic changes in the X-ray pulse profile that track the 35-day cycle.

The most viable alternative is therefore the precession of a warped, inclined accretion disc, conceivably driven by tidal torques from the companion star (J. I. Katz 1973; D. Gerend & P. E. Boynton 1976; J. A. Pettersson 1977; D. M. Scott et al. 2000; D. A. Leahy et al. 2020; D. Leahy & S. Frost 2025). The modulation of the polarization signal could then arise from scattering within the accretion-disc corona (D. Leahy & S. Frost 2025, but see J. Heyl et al. 2024) although whether the

IXPE data can be fitted with such a model remains to be demonstrated.

6. CONCLUSIONS

We have examined the implications of long-lived free precession of the neutron star in Her X-1 for the dynamics of the internal superfluid. Sustaining precession over the ~ 50 yr observational baseline requires an extremely weak effective coupling between the superfluid and the normal component, corresponding to a drag parameter $\mathcal{R} \lesssim 10^{-2}$.

This result carries important consequences for standard models of pulsar glitches. In particular, the differential rotation that develops during the quasi-steady precession phase is many orders of magnitude smaller than the critical lag expected for vortex unpinning, and is also far below the fractional spin changes observed in glitches. If the superfluid responsible for storing angular momentum in glitching pulsars behaves similarly in accreting neutron stars such as Her X-1, our results suggest that either the region involved in glitches is largely decoupled from the precessing component, or the effective pinning and drag depend sensitively on the dynamical state of the star.

Finally, we note that alternative explanations for the 35-day cycle in Her X-1 remain possible, and further observational and theoretical work will be required to determine whether they can account for the observed polarization properties.

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APPENDIX

A. EFFECT OF THE CRUST-CORE COUPLING

Within the main body of the paper, we implicitly assume that the star's crust is tightly coupled to the core. Thus I_0 is the sum of the crust and core's moments of inertia: $I_0 = I_{\text{cr}} + I_{\text{co}} = I_{\text{cr}}(1 + \kappa_{\text{co}})$, where $\kappa_{\text{co}} \sim 10^2$ for neutron stars. However, in reality, the crust-core coupling, even being strong, is not perfectly tight. Following A. Alpar & H. Ogelman (1987), one may assume that the crust is subject to the torque from the core

$$N_{\text{co}} = I_{\text{co}} \tau_{\text{co}}^{-1} (\Omega_{\text{co}} - \Omega_{\text{n}}), \quad (\text{A1})$$

where Ω_{co} is the core angular velocity and τ_{co} is the coupling time scale.

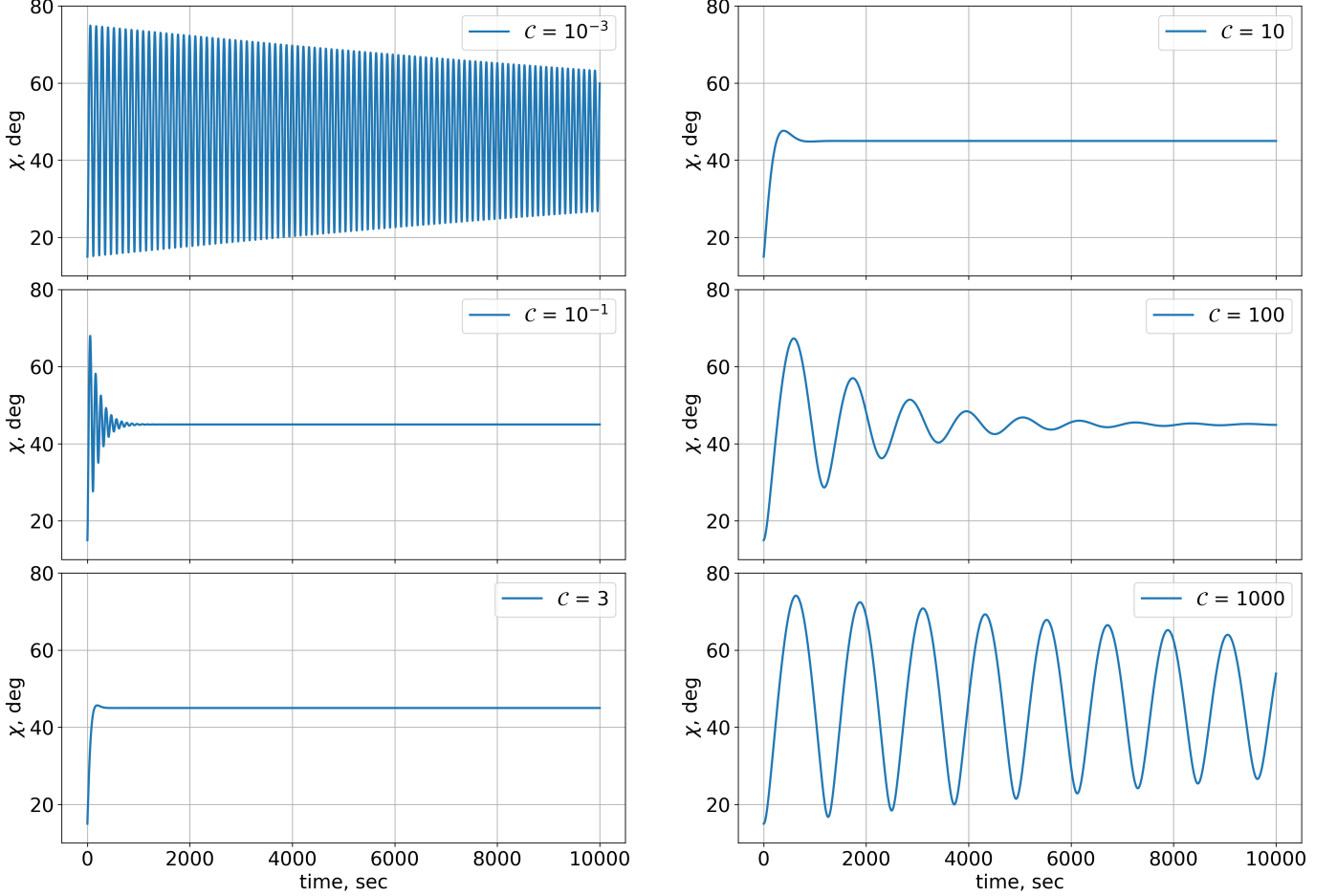


Figure 7. Evolution of the magnetic angle in a toy model including a strongly deformed crust ($\epsilon_{\text{cr}} = 10^{-2}$) and a massive core ($\kappa_{\text{co}} = 10$) coupled to it. The spin period of the star is $P_0 = 1$ s. The rotational evolution for different values of $\mathcal{C}$ is shown and is fully consistent with the theoretical predictions. For weak coupling ($\mathcal{C} = 10^{-3}$), the star exhibits long-lived precession with period $\sim 2P_0/\epsilon_{\text{cr}} \sim 200$ s, for an initial tilt angle $\theta_0 = 60^\circ$, and with the magnetic axis inclined by 45° with respect to the symmetry axis. For strong coupling ($\mathcal{C} = 10^3$), the precession period is longer by a factor κ_{co} , reaching ≈ 2000 s. In the intermediate regime, precession damps rapidly, particularly near $\mathcal{C} = 3 \approx \sqrt{1 + \kappa_{\text{co}}}$.

Consider the effect of imperfect crust–core coupling within a toy rotational model that neglects both the crustal superfluid and any external torques. In particular, we let the crust evolve according to

$$\dot{\boldsymbol{\Omega}}_{\text{n}} + \dot{\boldsymbol{\omega}} = (\boldsymbol{\omega} \times \boldsymbol{\Omega}_{\text{n}}) + \kappa_{\text{co}} \tau_{\text{co}}^{-1} (\boldsymbol{\Omega}_{\text{co}} - \boldsymbol{\Omega}_{\text{n}}), \quad (\text{A2})$$

where $\boldsymbol{\omega} = (0, 0, \epsilon_{\text{cr}} \Omega_{\text{n}} \cos \theta)^T$, with $\epsilon_{\text{cr}} = \Delta I_{\text{cr}}/I_{\text{cr}}$ characterizing the crust deformation. Note that ϵ_{cr} does not represent the asymmetry of the total stellar moment of inertia, unlike the parameter ϵ used in the main body of the paper. As will become clear later, ϵ_{cr} can be significantly larger than ϵ .

The rotational evolution of the core is given by

$$\dot{\boldsymbol{\Omega}}_{\text{co}} = (\boldsymbol{\Omega}_{\text{co}} \times \boldsymbol{\Omega}_{\text{n}}) - \tau_{\text{co}}^{-1} (\boldsymbol{\Omega}_{\text{co}} - \boldsymbol{\Omega}_{\text{n}}), \quad (\text{A3})$$

in the frame pinned to the crust. Let

$$\mathcal{C} = \frac{1 + \kappa_{\text{co}}}{\Omega \tau_{\text{co}}} \quad (\text{A4})$$

be the dimensionless coupling parameter. In the limit $\mathcal{C} \rightarrow \infty$, the crust and core are perfectly coupled and rotate as a single rigid body. Conversely, for $\mathcal{C} = 0$, the core dynamics has no influence on the crust. Since the second term on the right-hand side of Eqs. (A2)–(A3) is dissipative, it plays a role analogous to that of the crustal superfluid discussed in

the previous sections. Specifically, on a short initial timescale ($\sim \tau_{\text{co}}/\kappa_{\text{co}}$), the crust and core spin axes relax toward a quasi-equilibrium misalignment angle

$$\gamma_{\text{cr-co,eq}} = \epsilon_{\text{cr}} \frac{\sin \theta \cos \theta}{\sqrt{1 + \mathcal{C}^2}}. \quad (\text{A5})$$

After this transient stage, both components evolve jointly.

By rewriting Eqs. (A2)–(A3) and assuming $\gamma_{\text{cr-co}} \ll 1$, one finds that in the equilibrium state the crust precesses about the symmetry axis with angular velocity

$$\frac{d\varphi}{dt} = \omega_{\text{cr}} \frac{1 + (1 + \kappa_{\text{co}})^{-1} \mathcal{C}^2}{1 + \mathcal{C}^2}, \quad (\text{A6})$$

while the tilt angle decays at the rate

$$\frac{d\theta}{dt} = -\epsilon_{\text{cr}} \kappa_{\text{co}} \tau_{\text{co}}^{-1} \frac{\sin \theta \cos \theta}{1 + \mathcal{C}^2}. \quad (\text{A7})$$

Equivalently, in terms of the number of precession cycles,

$$\frac{d\theta}{\dot{\varphi} dt} = -\frac{\kappa_{\text{co}} \mathcal{C}}{1 + \kappa_{\text{co}} + \mathcal{C}^2} \sin \theta, \quad (\text{A8})$$

which is independent of ϵ_{cr} .

This result implies that in the no-coupling regime ($\mathcal{C} = 0$), the crust precesses at the rate ω_{cr} without any damping. This is a trivial result. On the other hand, in the limit of absolute coupling ($\mathcal{C} \rightarrow \infty$), the precession becomes significantly slower:

$$\dot{\varphi} = \frac{\omega_{\text{cr}}}{1 + \kappa_{\text{co}}}, \quad (\text{A9})$$

as expected, since in this case the star behaves as a single rigid body with total moment of inertia $I_{\text{cr}}(1 + \kappa_{\text{co}})$. In this limit, the precession is also undamped, $\dot{\theta} = 0$.

However, for finite coupling, the precession period exceeds $2\pi/\omega_{\text{cr}}$, and the spin axis relaxes toward either the symmetry axis ($\epsilon_{\text{cr}} > 0$) or the symmetry equator ($\epsilon_{\text{cr}} < 0$) within a finite number of cycles. The fastest damping occurs for $\mathcal{C} = \sqrt{1 + \kappa_{\text{co}}}$, corresponding to

$$\tau_{\text{co}} = \frac{P\sqrt{1 + \kappa_{\text{co}}}}{2\pi\kappa_{\text{co}}} \approx \frac{P}{63}, \quad (\text{A10})$$

for $\kappa_{\text{co}} = 10^2$. However, A. Alpar & H. Ogelman (1987) estimate $\Omega\tau_{\text{co}} \sim 10^2 - 10^4$ for realistic neutron stars, which implies $\mathcal{C} \sim 10^{-2} - 1$. For the upper end of this range, the crust–core coupling damps the precession on a timescale of order one cycle. In contrast, if $\mathcal{C} \sim 10^{-2}$, the precession can survive for hundreds of cycles with period $\approx 2\pi/\omega_{\text{cr}}$.

To illustrate the effect of crust–core coupling, we obtain a full numerical solution of the system (A2)–(A3) for a biaxial star with the following parameters: $\epsilon_{\text{cr}} = 10^{-2}$, $P_0 = 1$ s, and $\theta_0 = 60^\circ$. The magnetic axis is assumed to be tilted by 45° with respect to the symmetry axis, corresponding to an initial magnetic inclination $\chi_0 = 15^\circ$. The ratio of the core to the crust moments of inertia in this toy model is set to $\kappa_{\text{co}} = 10$.

In Fig. 7 we show the simulated evolution of the magnetic angle in this model for a range of $\mathcal{C}$ values from 10^{-3} to 10^3 . The results are fully consistent with the theoretical predictions. For very weak coupling ($\mathcal{C} = 10^{-3}$), the system exhibits long-lived precession with period $2\pi/\omega_{\text{cr}} \approx 200$ s. However, a slow decay of the precession amplitude is observed. For larger values of $\mathcal{C}$, damping of the precession becomes more significant, reaching its strongest regime near $\mathcal{C} \approx 3 \sim \sqrt{1 + \kappa_{\text{co}}}$. Beyond this regime, strong coupling leads again to long-lived precession, but with a reduced frequency $\omega_{\text{cr}}/(1 + \kappa_{\text{co}})$.

B. STEADY-STATE ROTATION PARAMETERS

The evolution of $\delta\mathbf{\Omega} = \mathbf{\Omega}_s - \mathbf{\Omega}_n$ in the linear approximation, Eq. (22), reads:

$$\frac{d(\delta\mathbf{\Omega})}{dt} = -(\boldsymbol{\omega} \times \mathbf{\Omega}_n) - \frac{\mathbf{N}_{\text{ext}}}{I_0} - (\mathbf{\Omega}_{\text{Sh}} \times \delta\mathbf{\Omega}) - \frac{\delta\mathbf{\Omega} + (\Omega_s/\Omega_n - 1)\mathbf{\Omega}_n}{\tau_d}. \quad (\text{B11})$$

It is convenient to transform to the basis vectors $\mathbf{e}_\pm = \mathbf{e}_\varphi \pm i\mathbf{e}_\theta$. Then $\mathbf{e}_\Omega \times \mathbf{e}_\pm = \pm i\mathbf{e}_\pm$, and $\delta\mathbf{\Omega} = \delta\mathcal{W}_\perp^- \mathbf{e}_+ + \delta\mathcal{W}_\perp^+ \mathbf{e}_-$, where $2\delta\mathcal{W}_\perp^\pm = \Omega_n(\varphi_s - \varphi) \sin \theta \pm i\Omega_n(\theta_s - \theta)$. For the parallel component, Eq. (B11) reduces to

$$\frac{d}{dt}(\delta\Omega_\parallel) + 2\tau_d^{-1}\delta\Omega_\parallel = -\Omega_n^2 n_\Omega, \quad (\text{B12})$$

and for the projection of the orthogonal component on $\mathbf{e}_\mp$, to

$$\frac{d}{dt}(\delta\mathcal{W}_\perp^\pm) + \mathcal{W}_{\text{Sh}}\delta\mathcal{W}_\perp^\pm = -\Omega_n^2 \frac{\mathcal{N}_\perp^\pm}{2}, \quad (\text{B13})$$

where $\mathcal{W}_{\text{Sh}} \equiv \tau_d^{-1} - i\Omega_{\text{Sh}}$ and $\mathcal{N}_\perp^\pm \equiv (\epsilon \sin \theta \cos \theta + n_\varphi) \pm in_\theta$ are complex representations of the Shaham's frequency, and the projections of the torque on $\mathbf{e}_\mp$, respectively. The dimensionless torque components n_θ and n_φ are defined by Eq. (29) above. Thus, $\tau_d = (1 + \mathcal{R}^2)/(\Omega_n \mathcal{R}(1 + \kappa_s))$ is simply the decay time of the rapid Shaham's mode.

The external torque components change on a time-scale given by the minimum of two: the precession damping time τ_p and the spin-down time. However, both are longer than τ_d provided that $\kappa_s/|\epsilon| \ll 1$. Therefore, the right-hand sides of Eqs. (B12) and (B13) can be treated as constants evolving adiabatically. Their solutions are

$$\delta\Omega_\parallel(t) = -\frac{1}{2}\Omega_n^2 n_\Omega \tau_d \left(1 - e^{-t/2\tau_d}\right) + \delta\Omega_{\parallel,0} e^{-t/2\tau_d}, \quad (\text{B14})$$

and

$$\delta\mathcal{W}_\perp^\pm(t) = -\Omega_n^2 \frac{\mathcal{N}_\perp^\pm \mathcal{W}_{\text{Sh}}^*}{2|\mathcal{W}_{\text{Sh}}|^2} (1 - e^{-\mathcal{W}_{\text{Sh}} t}) + \delta\mathcal{W}_{\perp,0}^\pm e^{-\mathcal{W}_{\text{Sh}} t} \quad (\text{B15})$$

respectively. These solutions show that if τ_d is sufficiently short, the exponential terms decay rapidly, and the components of the lag approach slowly-evolving values. Consequently, both $\delta\dot{\Omega}_\parallel$ and $\delta\dot{\mathcal{W}}_\perp^\pm$ become close to zero, corresponding to a quasi-steady-state evolution. The corresponding lag components are

$$\delta\Omega_{\parallel,\text{eq}} = -\Omega_n^2 n_\Omega \tau_d / 2, \quad (\text{B16})$$

and

$$\gamma_{\text{eq}}^2 = \delta\mathcal{W}_{\perp,\text{eq}}^+ \delta\mathcal{W}_{\perp,\text{eq}}^- = \Omega_n^4 \frac{\mathcal{N}_\perp^+ \mathcal{N}_\perp^-}{|\mathcal{W}_{\text{Sh}}|^2}. \quad (\text{B17})$$

These expressions coincide with those given earlier in Eqs. (28) and (27), respectively. The components of the latter along the polar and azimuthal directions are

$$\delta\theta_{\text{eq}} = -\frac{1 - \kappa_s \mathcal{R}^2}{1 + \kappa_s^2 \mathcal{R}^2} (n_\varphi + \epsilon \sin \theta \cos \theta) - \frac{(1 + \kappa_s) \mathcal{R}}{1 + \kappa_s^2 \mathcal{R}^2} n_\theta, \quad (\text{B18})$$

and

$$(\delta\varphi \sin \theta)_{\text{eq}} = \frac{1 - \kappa_s \mathcal{R}^2}{1 + \kappa_s^2 \mathcal{R}^2} n_\theta - \frac{(1 + \kappa_s) \mathcal{R}}{1 + \kappa_s^2 \mathcal{R}^2} (n_\varphi + \epsilon \sin \theta \cos \theta) \quad (\text{B19})$$

respectively.

Following the same approach, the precessional evolution can be treated by introducing the complex quantity $\mathcal{W}_\perp^+ = \Omega_n \varphi \sin \theta + i\Omega_n \theta$, so that

$$\frac{d}{dt} \mathcal{W}_\perp^+ = \Omega_n^2 \mathcal{N}_\perp^+ + \Omega_n \frac{\kappa_s \mathcal{R}}{1 + \mathcal{R}^2} (1 + i\mathcal{R}) \delta\mathcal{W}_\perp^+. \quad (\text{B20})$$

If in this equation $\delta\mathcal{W}_\perp^+ = \delta\mathcal{W}_{\perp,\text{eq}}^+$, the solution coincides with those for $\dot{\varphi}$ and $\dot{\theta}$ obtained in the main text, Eqs. (32) and (31), respectively.

C. HERCULES X-1 EXTERNAL TORQUE

As noted in Sect. 4 above, the neutron star in the Her X-1 system is subject to the influence of an external torque, which can be considered as a sum of several components:

$$\mathbf{N}_{\text{ext,HX1}} = \mathbf{N}_{\text{acc}} + \mathbf{N}_{\text{mag}} + \mathbf{N}_{\text{psr}} + \mathbf{N}_{\text{warp}} + \mathbf{N}_{\text{prec}}. \quad (\text{C21})$$

In this Appendix, we provide details of these torques as they have been adopted in the numerical calculations. The first component in the equation above represents the accretion spin-up:

$$\mathbf{N}_{\text{acc}} = \dot{M} f(\cos \xi) \sqrt{GM_{\text{NS}} r_{\text{m}}} \mathbf{d} = N_{\text{acc},0} f(\cos \xi) \mathbf{d}, \quad (\text{C22})$$

where $\dot{M}$ is the accretion rate, M_{NS} is the star mass, $\mathbf{d}$ is the disc angular momentum axis ($|\mathbf{d}| = 1$),

$$r_{\text{m}} = \frac{1}{2} \left(\frac{\mu^4}{2GM_{\text{NS}}\dot{M}^2} \right)^{1/7} \quad (\text{C23})$$

is the inner magnetospheric radius and

$$f(\cos \xi) = A(\eta, \alpha, \chi)(1 - \eta \cos^2 \xi) \quad (\text{C24})$$

is the modulation factor depending on the instant angle between the star's magnetic $\mathbf{m}$ and disc axes: $\xi = \cos^{-1}(\mathbf{m} \cdot \mathbf{d})$. The normalizing coefficient $A(\eta, \alpha, \chi)$ depend on the magnetic inclination χ , spin-disc inclination $\alpha = \cos^{-1}(\mathbf{e}_{\Omega} \cdot \mathbf{d})$ and modulation parameter $\eta = 0..1$ (see [A. Biryukov & P. Abolmasov 2021](#), for details) as

$$A = \left[1 - \frac{\eta}{2} (\sin^2 \chi \sin^2 \alpha + 2 \cos^2 \chi \cos^2 \alpha) \right]^{-1}. \quad (\text{C25})$$

The second term in Eq. (C21) describes the spin-down due to the interaction of the disc by the stellar magnetic field:

$$\mathbf{N}_{\text{mag}} = -\frac{\mu^2}{3r_{\text{c}}^3} \mathbf{e}_{\Omega} = N_{\text{mag}} \mathbf{e}_{\Omega}, \quad (\text{C26})$$

where

$$r_{\text{c}} = \left(\frac{GM_{\text{NS}}}{\Omega^2} \right)^{1/3} \quad (\text{C27})$$

is the corotation radius.

The third term represents pulsar losses:

$$\mathbf{N}_{\text{psr}} = N_{\text{psr},0} (1 + \sin^2 \chi) \mathbf{e}_{\Omega} + N_{\text{psr},0} \cos \chi ((\mathbf{m} \times \mathbf{e}_{\Omega}) \times \mathbf{e}_{\Omega}) + \frac{N_{\text{psr},0}}{3} \left(\frac{r_{\text{LC}}}{r_{\text{NS}}} \right) \cos \chi (\mathbf{m} \times \mathbf{e}_{\Omega}), \quad (\text{C28})$$

and depends on the instant direction of the crust spin axis, magnetic angle χ , and scales as $N_{\text{psr},0} = -\mu^2/r_{\text{LC}}^3$, where $r_{\text{LC}} = c/\Omega$ is the light cylinder radius ([P. Abolmasov et al. 2024](#)). At this point, we ignore pulsar losses enhancement due to the extra opening of the magnetic field lines by the disc ([K. Parfrey et al. 2016](#)), but take into account the so-called ‘‘anomalous’’ part of the pulsar torque. The latter scales as $(r_{\text{LC}}/r_{\text{NS}})N_{\text{psr},0}$ and hence is much stronger than standard pulsar losses. For this component, we adopted the equation found by [V. S. Beskin & A. A. Zheltoukhov \(2014\)](#) in the assumption that the star's internal magnetic field is uniform and aligned with the magnetic axis.

The two final terms account for backreaction from the warping and precession of the tilted disc:

$$\mathbf{N}_{\text{warp}} = \frac{4\sqrt{2}}{3} N_{\text{acc},0} \cos^2 \chi \cos \alpha (\mathbf{d} \times (\mathbf{e}_{\Omega} \times \mathbf{d})) \quad (\text{C29})$$

and

$$\mathbf{N}_{\text{prec}} = \frac{32\sqrt{2}}{3\pi} N_{\text{acc},0} \sin^2 \chi \cos \alpha (\mathbf{e}_{\Omega} \times \mathbf{d}) \quad (\text{C30})$$

respectively ([D. Lai 1999](#); [D. Lai et al. 2011](#)).

Until the disc angular momentum is aligned to the star spin axis ($\alpha = 0$), the fundamental equations (15) and (16) must be supplemented with another one, describing the evolution of α :

$$I_0 \Omega \frac{d\alpha}{dt} = -\mathbf{N}_{\text{ext,HX1}} \cdot \mathbf{l}, \quad (\text{C31})$$

where $\mathbf{l}$ is the unit vector that always lies within the $\mathbf{e}_{\Omega} - \mathbf{d}$ plane, orthogonal to the spin axis, and makes an acute angle with $\mathbf{d}$. However, $\mathbf{N}_{\text{ext,HX1}}$ includes rapidly oscillating torques $\mathbf{N}_{\text{acc}}$, $\mathbf{N}_{\text{warp}}$, and $\mathbf{N}_{\text{prec}}$ that rotate around $\mathbf{\Omega}$ with on the spin period time scale, when considered in the principal frame. Thus, following the approach used by [A. Biryukov & P. Abolmasov \(2021\)](#), we hereafter average the net torque projections over the spin period to obtain analytic expressions for the components of $\mathbf{N}_{\text{ext,HX1}}$. In particular Eq.(C31) then reads as

$$\left\langle I_0 \Omega \frac{d\alpha}{dt} \right\rangle = -\langle \mathbf{N}_{\text{ext,HX1}} \cdot \mathbf{l} \rangle = -N_{\text{acc},0} \sin \alpha \left(1 - \frac{4\sqrt{2}}{3} \cos^2 \alpha \cos^2 \chi \right). \quad (\text{C32})$$

Since the expression in the parentheses on the RHS of this equation is capable of changing sign, a stationary value of α can exist, in principle. Spin-disk alignment reaches it on the accretion time scale and then continues to evolve on the $|\chi/\dot{\chi}|$ time scale (see Fig. 4 from D. Lai et al. (2011)). The detailed analysis of that is beyond the scope of our paper.

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